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Willingness to Pay for Carbon Mitigation: Field Evidence from the Market for Carbon Offsets

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1 Big picture

- **Question.** What does real consumer demand for voluntary carbon offsets reveal about households' willingness to pay for carbon mitigation?
- **Empirical setting.** The paper runs a large-scale natural field experiment inside the online shop of a major German grocery delivery company. Consumers encounter an optional carbon offset in the shopping basket and make real purchase decisions.
- **Baseline product.** The baseline offset compensates the average emissions of one delivery, 2.4 kg of CO₂, at a price of 24 cents. This corresponds to 10 cents per kg, or 100 EUR/tCO₂.
- **Experimental variation.** The experiment randomizes (i) whether the price is subsidized by 50% or 75%, (ii) whether the quantity of carbon offset is matched upward by 100% or 300%, and (iii) whether consumers are explicitly told that the firm pays the remaining offset cost.
- **Main revealed-preference finding.** Consumers in the standard treatments are price-elastic but fully impact-inelastic: lower prices raise offset purchases, but more carbon mitigation does not. This implies a revealed WTP for carbon mitigation close to zero when demand is interpreted through the standard model.
- **Role of firm contribution information.** When the firm makes its own contribution salient, consumers become more price-elastic and start responding to impact. The estimated WTP rises to about 16 EUR/tCO₂ in the information treatment.
- **Behavioral interpretation.** Demand for offsets appears to combine warm glow, inattention to scope, and fairness/equity preferences about cost sharing between consumers and the firm. Consumers buy offsets partly because buying the offset feels like doing something good, not because they always value the quantity of carbon mitigated.
- **Cost-effectiveness result.** Quantity matches are more cost-effective than price subsidies. Subsidies mostly transfer money to inframarginal consumers, while matches increase carbon mitigation for both marginal and inframarginal consumers.
- **Survey comparison.** A complementary stated-preference survey yields a mean WTP of 238 EUR/tCO₂, far above the field-experiment estimates. This provides a sharp warning about hypothetical bias in climate WTP surveys.
- **Economic takeaway.** Voluntary carbon-offset demand does not automatically reveal strong intrinsic valuation of carbon mitigation. The way the firm frames its contribution is central: it can activate fairness preferences and attention to impact, but revealed WTP remains much smaller than stated WTP.

2 Data and measurement

2.1 Online supermarket field setting

- The experiment is embedded in a real online supermarket, so consumers are not recruited to a lab and do not know they are part of an experiment.
- The paper observes more than 250,000 consumers, more than 400,000 website visits, and 108,478 orders over the experimental period.
- Treatment assignment occurs when consumers visit the website and is tied to HTTP-cookie identifiers.
- The offset appears at checkout, next to the consumer's shopping basket, which makes the setting resemble real-world add-on offset offers for flights, deliveries, and other services.
- The experiment focuses on actual purchases, not intentions, survey responses, or hypothetical choices.

Interpretation: The market setting is a major contribution: consumers face a real price, make a real decision, and spend real money. This is why the results are informative about revealed preferences rather than merely stated environmental attitudes.

2.2 Baseline offset product

- The baseline offset compensates 2.4 kg of CO₂, which the firm calculates as the average emissions of one delivery.
- The baseline price is 24 cents, which corresponds to 10 cents per kg of CO₂, or 100 EUR per ton.
- The offset is directed to a reforestation project that plants trees to compensate emissions.
- The offset is added by ticking a checkbox, so consumers must actively opt in.
- Consumers are told that the 2.4 kg amount corresponds to the average emissions of one delivery, providing a reference point for interpreting the quantity.

Interpretation: The baseline product is small-stakes but naturalistic. The small price helps identify behavior at the point where offset add-ons typically appear in consumer markets.

2.3 Treatment arms

The treatment design has ten groups:

- **Control:** no offset is offered.
- **Baseline:** 2.4 kg of CO₂ for 24 cents.
- **Price subsidy treatments:** consumers pay 12 cents or 6 cents for the same 2.4 kg offset.
- **Quantity match treatments:** consumers pay 24 cents but offset 4.8 kg or 9.6 kg.
- **Information versions:** for each subsidy and match, the firm also explicitly tells consumers that it pays the remaining cost of the offset.

Interpretation: The key symmetry is that a 50% subsidy and a 100% match both split the offset cost 50:50 between consumer and firm. A 75% subsidy and a 300% match both split costs 25:75. This allows the paper to compare price reductions and quantity matches holding the underlying cost-sharing structure comparable.

2.4 Standard versus information treatments

- In **STANDARD**, consumers see only a lower price or larger offset quantity.
- In **INFORMATION**, consumers are explicitly informed that the firm finances the subsidy or the match.
- The information treatment can affect behavior through several channels: fairness preferences, attention to scope, trust in the project, perceived firm responsibility, or beliefs about offset quality.
- The main interpretation is that **INFORMATION** makes the firm's own contribution salient and thereby changes the psychology of the offset purchase.

Interpretation: The information cross-randomization is critical because a subsidy or match by itself is not purely a price or quantity treatment. It also changes the implicit cost sharing between consumer and firm. The information treatment makes that cost sharing visible.

2.5 Post-purchase survey

After placing an order, eligible consumers answer two belief questions:

- How large are the negative environmental consequences of the delivery if emissions are not compensated?
- How effective is the offset program in reducing these negative consequences?

Interpretation: The post-purchase survey helps diagnose mechanisms. If consumers respond to low prices because they infer that the environmental damage is small or the offset is ineffective, the survey can reveal changes in beliefs caused by treatments.

2.6 Complementary email survey

- Several months after the field experiment, the firm invites customers to complete an opinion survey.
- The survey elicits hypothetical WTP for offsetting 2.4, 4.8, or 9.6 kg of CO₂.
- It randomizes an education treatment about carbon emissions and an information treatment about firm matching.
- It also asks what share of offset costs the firm should pay and whether respondents support a carbon tax.

Interpretation: The survey is useful for comparing revealed and stated preferences in the same consumer environment. Its main lesson is that stated WTP is much larger than revealed WTP.

3 Models and empirical specifications

3.1 Treatment-effect regressions

The reduced-form treatment effects compare offset purchase probabilities across randomized groups. The within-subject panel specification can be written conceptually as

$$D_{it} = \alpha_i + \mu_t + \beta_1 Subsidy_{it} + \beta_2 Match_{it} + \beta_3 Information_{it} + \beta_4 (Subsidy_{it} \times Information_{it}) + \beta_5 (Match_{it} \times Information_{it}) + \varepsilon_{it}.$$

Here D_{it} is an indicator that consumer i buys the offset during visit/order t .

Interpretation: Between-subject estimates use the first treatment exposure and exploit randomization across consumers. Within-subject estimates use repeated consumer visits and ask whether the same consumer behaves differently when assigned to different treatments over time.

3.2 Baseline random utility model

Consumer i chooses between buying the offset and an outside option. The outside option is normalized to zero. The offset compensates γ_i units of carbon and costs price p_i . Baseline utility is

$$u_i = \omega + \beta\gamma_i + \eta p_i + \epsilon_i.$$

- ω is warm glow from buying the offset, independent of impact.
- β is utility from carbon mitigation.
- $\eta < 0$ is marginal utility of price.
- ϵ_i captures idiosyncratic tastes.

Interpretation: This model separates the act of giving from the amount of carbon mitigated. Warm glow can generate demand even if $\beta = 0$.

3.3 WTP from demand elasticities

The consumer buys if $u_i \geq 0$. Aggregate demand is $D = 1 - G(\xi)$, where $\xi = -\beta\gamma - \eta p - \omega$. The derivatives are

$$\frac{\partial D}{\partial p} = \eta g(\xi), \quad \frac{\partial D}{\partial \gamma} = \beta g(\xi).$$

Therefore,

$$\text{WTP} = -\frac{\beta}{\eta} = -\frac{\partial D / \partial \gamma}{\partial D / \partial p}.$$

Interpretation: WTP is identified by the ratio of the quantity response to the price response. This is why the design varies both offset price and offset quantity.

3.4 Linear approximation

The baseline utility parameters can be estimated with

$$D_i = \alpha + \tau\gamma_i + \zeta p_i + \xi_i \approx -\frac{\omega}{\eta}\zeta - \frac{\beta}{\eta}\zeta\gamma_i + \zeta p_i + \xi_i.$$

Interpretation: The coefficient on carbon quantity identifies the valuation of carbon mitigation relative to the price coefficient. If demand does not respond to quantity, WTP is zero even if demand responds to price.

3.5 Behavioral model: inattention, scope, and fairness

The extended model allows consumers to be attentive or inattentive to impact and to care about cost sharing between the consumer and the firm:

$$u_i = \omega + \theta\beta\gamma_i + \eta p_i + F(p_i, s_i, I_i) + \epsilon_i.$$

Here s_i is the firm contribution and I_i is an indicator that the contribution is made salient. The estimated linear form is

$$D_i = \alpha + \tau\gamma_i + \zeta p_i + \psi \max\{s_i I_i - p_i, 0\} + \varphi \max\{p_i - s_i I_i, 0\} + \xi_i.$$

Interpretation: This extension recognizes that consumers may not process kg of carbon as a salient attribute unless the experimental design highlights impact, and that consumers may value the firm paying at least as much as they do.

3.6 GMM estimation with between- and within-subject moments

The paper estimates behavioral parameters jointly using GMM. Between-subject moments identify warm glow and fairness terms, while within-subject moments identify WTP and attenuation after controlling for consumer and time fixed effects.

$$D_i^b = \zeta p_i - \frac{\omega}{\eta}\zeta - \frac{\phi}{\eta}\zeta \max\{s_i I_i - p_i, 0\} - \frac{\rho}{\eta}\zeta \max\{p_i - s_i I_i, 0\} + \xi_i^b,$$

$$D_{it}^w = \left(\zeta p_{it} - \frac{\beta}{\eta}\zeta\gamma_{it} - \frac{\phi}{\eta}\zeta \max\{s_{it} I_{it} - p_{it}, 0\} - \frac{\rho}{\eta}\zeta \max\{p_{it} - s_{it} I_{it}, 0\} \right) \Omega + \lambda_i + \mu_t + \xi_{it}^w.$$

Interpretation: The key purpose of the GMM model is to combine the clean between-subject randomization with the stronger within-subject scope comparison. The attenuation parameter Ω accounts for weaker effects after repeated exposure.

3.7 Cost-effectiveness metric

The paper defines intervention cost effectiveness as

$$\text{Cost effectiveness} = \frac{\Delta \text{kg of compensated carbon relative to baseline}}{\text{EUR spent by firm}}.$$

Interpretation: This converts demand responses into an implementation metric for firms. A subsidy is cost-effective only if marginal demand responds enough. A match is cost-effective because it raises the carbon quantity for everyone who buys, including inframarginal buyers.

3.8 Survey WTP regressions

The email survey estimates OLS regressions of stated WTP on quantity increases, education treatments, firm contribution information, and carbon-tax support.

Interpretation: The survey tests whether hypothetical WTP responds to scope and firm contribution in the same way as revealed WTP. The answer is mostly no: stated WTP is very high, and the firm contribution information does not reproduce the field-experiment pattern.

4 Identification and empirical design

4.1 Core identification problem

The paper must identify how much consumers value carbon mitigation, not just whether they like buying offsets. An offset purchase can reflect at least five components:

- intrinsic valuation of reduced emissions;
- warm glow from the act of giving;
- responsiveness to price;
- attention or inattention to the amount of carbon compensated;
- fairness preferences about whether the firm shares costs.

Interpretation: A simple offset purchase rate is not a WTP measure. Without impact variation, the act of buying could be mistaken for valuing carbon mitigation.

4.2 Why randomizing price and quantity identifies WTP

- Price variation identifies the marginal utility of money.
- Quantity variation identifies whether demand responds to the amount of carbon mitigated.
- The ratio of quantity and price elasticities identifies WTP for carbon mitigation.
- Random assignment makes the variation orthogonal to consumer preferences and outside options.

Interpretation: This is the central identification logic. Consumers may have private outside options for carbon reduction, but randomization still identifies aggregate price and quantity responses within the shop.

4.3 Why the information treatment is necessary

- A subsidy changes price but also implies firm cost sharing.
- A match changes impact but also implies firm cost sharing.
- Consumers may respond to firm cost sharing, not only to price or carbon quantity.
- The information treatment makes firm cost sharing salient, allowing the paper to distinguish standard demand responses from demand responses shaped by fairness and firm engagement.

Interpretation: INFORMATION is not merely a label. It changes the meaning of the transaction: consumers learn the firm is participating in the offset.

4.4 Why within-subject variation matters

- Between-subject quantity variation shows consumers are impact-inelastic in STANDARD.
- Within-subject variation is more likely to make scope differences salient because the same consumer sees changing quantities over time.
- This helps separate true indifference from inattention or weak mental representation of impact.
- Subject fixed effects remove stable consumer-level tastes for environmental protection.

Interpretation: The within-subject design is used to diagnose whether zero revealed WTP in STANDARD means consumers do not value mitigation at all or simply fail to process the quantity dimension.

4.5 Randomization and sample balance

- Each of the ten groups contains roughly 25,000 consumers.
- Table 1 shows balance in website visits, purchase probabilities, expected travel times, and expected service times.
- The paper also reports that treatments do not affect the probability of placing an order at the shop.

Interpretation: This supports a causal interpretation of differences in offset demand: the treatments affect offset behavior without changing who buys groceries.

4.6 Identification threats

- **Outside options:** Consumers may offset emissions elsewhere. The design handles this by identifying WTP from randomized elasticities rather than from purchase levels alone.
- **Low-stakes extrapolation:** The offset costs only cents, so WTP may not extrapolate to large climate decisions.
- **Project trust:** Consumers may distrust offsets or infer quality from price. The post-purchase survey partially addresses beliefs about effectiveness.
- **Inattention:** Consumers may not understand kg of CO₂. The within-subject design and education survey help diagnose this.
- **Hypothetical bias:** The email survey shows stated WTP is much higher than revealed WTP.

Interpretation: The paper is strong on causal treatment effects within the shopping setting. It is more cautious about extrapolating the exact WTP number to high-stakes settings or other carbon products.

5 Section-by-section study guide

- **Introduction.** Explains why voluntary offsets are useful for measuring revealed environmental preferences and summarizes the key result: price response without impact response in STANDARD, but impact response when firm contribution is salient.
- **Experimental design.** Defines the ten treatment arms and explains the symmetry between subsidies and matches.
- **Results: between-subject effects.** Presents the main reduced-form demand patterns: price matters; impact does not unless the firm contribution is highlighted.
- **Post-purchase survey.** Tests whether information changes beliefs about environmental harm or offset effectiveness.
- **Within-subject effects.** Shows how demand changes when the same consumers face treatment variation over time.
- **Structural/behavioral model.** Maps treatment responses into WTP, warm glow, and fairness parameters.
- **Cost effectiveness.** Converts demand estimates into kg of carbon compensated per euro spent by the firm.
- **Email survey.** Compares revealed preferences with stated preferences and examines education, firm contribution, and carbon-tax support.
- **Conclusion.** Emphasizes that firms can influence voluntary mitigation, matches dominate subsidies in cost effectiveness, and stated WTP is highly inflated relative to revealed WTP.

6 Tables and figures

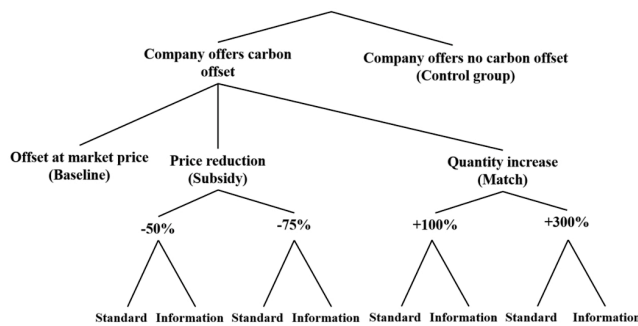
6.1 Figure 1: Experimental Design and Figure 2: Carbon Offset

Read:

- **Figure 1** lays out the randomization tree: consumers either see no offset, the baseline offset, a price subsidy, or a quantity match.
- Subsidies reduce the price by 50% or 75%; matches increase the amount of carbon mitigated by 100% or 300%.
- Each subsidy/match arm has a STANDARD version and an INFORMATION version.
- **Figure 2** shows the baseline carbon-offset product in the shopping basket: a checkbox, a 2.4 kg amount, and a 24-cent price.
- The baseline UI matters because this is the product consumers actually confront in the market.

Detailed interpretation: The design has two layers. The first layer varies the economic attribute: price or carbon quantity. The second layer varies salience of firm cost sharing. This is essential because a subsidy and a match both mechanically involve firm money, but consumers may or may not notice that the firm is participating. Figure 2 shows how small and add-on-like the decision is: the offset is embedded inside a grocery checkout, so the experiment captures behavior in a natural, low-attention environment rather than in a climate-policy survey.

Economic message: The experiment is designed to separate three forces that are normally confounded in offset markets: willingness to pay for mitigation, warm glow from giving, and willingness to share the offset burden with the firm.



(a) Figure 1: Experimental design

CO2 Compensation

☐ Yes, I would like to support environmental protection and offset 2.4kg CO2 for 24 Cents.

[Company] commits to pass on the entire amount to [project name]. An average delivery produces approx. 2.4kg CO2. The assessment basis for this calculation is the total fuel consumption divided by the number of deliveries.

(b) Figure 2: Carbon offset

6.2 Figure 3: Examples of Treatment Variation

Read:

- Panel A shows a STANDARD price subsidy: the consumer pays less for the same 2.4 kg offset.
- Panel B shows the INFORMATION version of the same subsidy: the consumer is told that the firm pays the remaining cost.
- Panel C shows a STANDARD quantity match: the consumer pays the same 24 cents but offsets more carbon.
- Panel D shows the INFORMATION version of the match: the consumer is told that the firm finances the additional offset quantity.

- The panels demonstrate that the same economic change can be framed either as an anonymous price/quantity change or as a joint consumer-firm contribution.

Detailed interpretation: This figure clarifies why the paper does not interpret subsidies and matches as purely mechanical price and quantity changes. In real consumer interfaces, wording matters. If the firm explicitly says it is contributing, consumers may interpret the purchase as a cooperative action, a fair cost split, or evidence that the offset is credible. Therefore, the treatment screens are not just implementation details; they are part of the causal design.

Economic message: The figure shows the mechanism behind the information treatment: firm participation turns the offset from a private add-on purchase into a shared responsibility frame.

A

CO2 Compensation

☐ Yes, I would like to support environmental protection and offset 2.4kg CO2 for 12 Cents.

[Company] commits to pass on the entire amount to [project named]. An average delivery problem appears. Using CO2. The assessment basis for this calculation is the total fuel consumption divided by the number of deliveries.

Price reduction by 50%

B

CO2 Compensation

☐ Yes, I would like to support environmental protection and offset 2.4kg CO2 for 12 Cents. The full compensation price for 2.4kg CO2 is 24 cents. [Company] pays the remaining 12 cents if I tick this box.

[Company] commits to pass on the entire amount to [project named]. An average delivery problem appears. Using CO2. The assessment basis for this calculation is the total fuel consumption divided by the number of deliveries.

Price reduction by 50%, with salient information

C

CO2 Compensation

☐ Yes, I would like to support environmental protection and offset 4.8kg CO2 for 24 Cents.

[Company] commits to pass on the entire amount to [project named]. An average delivery problem appears. Using CO2. The assessment basis for this calculation is the total fuel consumption divided by the number of deliveries.

Quantity increase by 100%

D

CO2 Compensation

☐ Yes, I would like to support environmental protection and offset 4.8kg CO2 for 24 Cents. The full compensation price for 4.8kg CO2 is 48 cents. [Company] pays the remaining 24 cents if I tick this box.

[Company] commits to pass on the entire amount to [project named]. An average delivery problem appears. Using CO2. The assessment basis for this calculation is the total fuel consumption divided by the number of deliveries.

Quantity increase by 100%, with salient information

Table 1
Summary table

Variable	Control	Baseline: 0.24kg at 2.4kg	-0.12kg	-0.18kg	-0.12kg, information
Number of website visits	1.593 (1.365)	1.596 (1.320)	1.585 (1.350)	1.588 (1.350)	1.602 (1.415)
Order (1= yes)	0.329 (0.470)	0.330 (0.470)	0.333 (0.471)	0.327 (0.469)	0.333 (0.471)
Offset (1= yes)	0.000 (0.000)	0.135 (0.342)	0.143 (0.350)	0.157 (0.364)	0.163 (0.369)
Expected travel time (in min)	14.558 (9.397)	14.566 (9.433)	14.468 (9.433)	14.509 (9.582)	14.561 (9.825)
Expected service time (in min)	7.201 (3.877)	7.380 (3.650)	7.384 (4.071)	7.282 (3.815)	7.286 (3.773)
N	25,560	25,427	25,604	25,556	25,643

Variable	-0.18kg, information	+2.4kg	+7.2kg	+2.4kg, information	+7.2kg, information
Number of website visits	1.381 (1.017)	1.391 (1.526)	1.388 (1.492)	1.392 (1.462)	1.398 (1.449)
Order (1= yes)	0.332 (0.471)	0.333 (0.472)	0.334 (0.472)	0.330 (0.470)	0.331 (0.471)
Offset (1= yes)	0.185 (0.388)	0.128 (0.334)	0.133 (0.330)	0.150 (0.375)	0.164 (0.371)
Expected travel time (in min)	14.525 (9.546)	14.442 (9.319)	14.428 (9.444)	14.470 (9.582)	14.685 (9.832)
Expected service time (in min)	7.371 (3.855)	7.334 (3.781)	7.385 (4.040)	7.285 (3.921)	7.302 (4.159)
N	25,575	25,564	25,762	25,642	25,189

Note: This table presents the mean of observable variables in different treatment conditions. Standard deviations are reported in parentheses.

6.3 Table 1: Summary Table

Read:

- The table reports observable means across the ten treatment groups.
- Each group contains about 25,000 consumers, supporting balance from random assignment.
- Website visits are about 1.6 per consumer across groups.
- Order probabilities are roughly 0.33 across groups.
- Expected travel time and service time are balanced, meaning the carbon-emission context of deliveries is similar across arms.
- Offsetting probabilities vary across offset treatment groups, which is the outcome the experiment is designed to move.

Detailed interpretation: The balance table is important because treatment assignment happens in a live online market. If treatments changed grocery-order behavior, differences in offset purchases could reflect selection into ordering rather than offset preferences. The table shows balanced pre-treatment and delivery characteristics. The control group has zero offset purchases by construction because it sees no offset, while positive offset rates in treatment arms reflect actual opt-in decisions.

Economic message: The table supports the credibility of the experiment: demand differences across arms can be interpreted as treatment effects on offset behavior rather than differences in who visits or orders.

6.4 Figure 4: Main Results

Read:

- Panel A plots offset purchase probabilities across price reductions and quantity increases.

- The baseline purchase rate is 13.5%.
- In STANDARD, reducing price by 12 and 18 cents raises purchase probability by 0.8 and 2.2 percentage points.
- In STANDARD, increasing offset quantity by 2.4 or 7.2 kg does not increase demand.
- In INFORMATION, price reductions raise demand more strongly: by 2.8 and 5 percentage points.
- In INFORMATION, quantity increases raise demand by 1.5 and 3 percentage points.
- Panel B compares cost effectiveness: kg of carbon compensated per euro spent by the firm.
- The dotted line is the market price benchmark of 10 kg/EUR.
- Quantity matches dominate subsidies in cost effectiveness because they increase carbon mitigation for inframarginal buyers as well as marginal buyers.

Detailed interpretation: Panel A is the central revealed-preference result. STANDARD consumers react to price but not to carbon quantity, so revealed WTP for the carbon-mitigation attribute is zero under the baseline interpretation. INFORMATION changes this: consumers now respond to impact. The pattern suggests that consumers are not simply indifferent to the environment; rather, the way the offset is framed determines whether they process impact. Panel B converts behavioral effects into firm policy guidance. Subsidies look appealing because they boost demand, but they spend money on all buyers who receive the lower price. Matches are more cost-effective because every buyer offsets more carbon.

Economic message: The figure delivers the paper’s core message: consumer offset demand is not automatically demand for carbon mitigation. It is shaped by salience, scope attention, and firm participation.

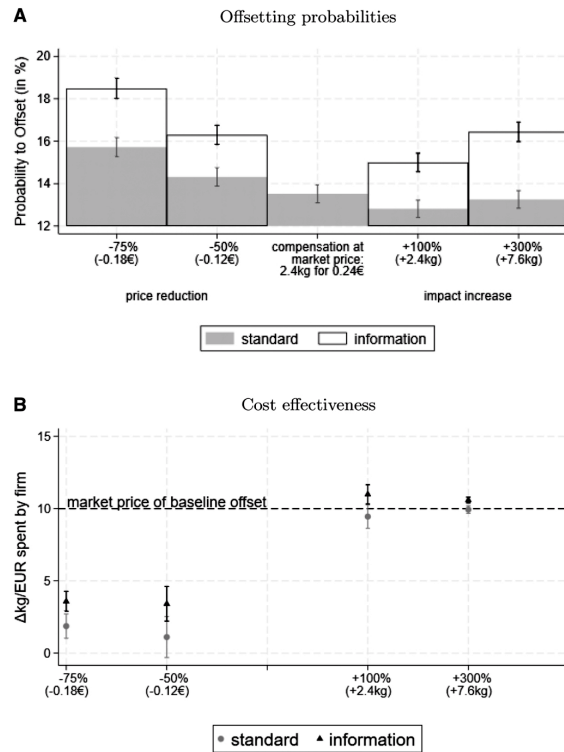


Figure 4: Main results

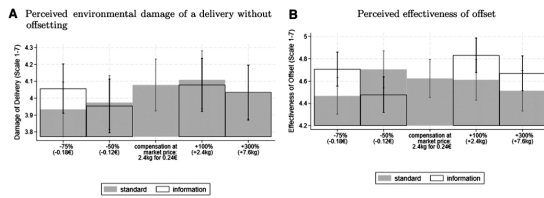
6.5 Figure 5: Post-Experimental Survey and Table 2: Between- and Within-Subject Treatment Effects

Read:

- **Figure 5** uses post-purchase beliefs to examine perceived environmental damage and offset effectiveness.
- Panel A asks whether consumers think an uncompensated delivery causes serious environmental harm.
- Panel B asks whether consumers think the offset program is effective.
- The figure suggests limited changes in perceived environmental damage across STANDARD and INFORMATION.
- There is suggestive evidence that INFORMATION may reduce the misperception that larger matches are ineffective.
- **Table 2** reports between- and within-subject treatment effects on offsetting probability.
- The subsidy effect is positive in between-subject variation and remains meaningful in within-subject variation.
- The match effect is weak in the first-visit between-subject estimates but becomes stronger in within-subject estimates.
- The information interaction is important because it shows that salience changes how consumers respond to price and quantity changes.

Detailed interpretation: Figure 5 helps rule in and rule out mechanisms. If INFORMATION mainly changed beliefs about the environmental harm of a delivery, panel A should move substantially; it does not. If INFORMATION makes consumers understand or trust the offset's effectiveness, panel B should move, and the figure provides some suggestive support. Table 2 complements Figure 4 by using regression structure and within-subject variation. The fact that consumers become more responsive to quantity when impact is made salient or repeated within-person suggests that scope inattention is a serious concern in measuring environmental preferences.

Economic message: The figure and table together indicate that demand is not governed by stable, fully informed carbon preferences. Attention and framing affect whether consumers respond to the amount of mitigation.



(a) Figure 5: Post-experimental survey

Table 2
Between- and within-subject treatment effects

	Offsetting Probability $\times 100$	
	(1)	(2)
subsidy	1.560*** (0.460)	2.579** (1.084)
× information	2.394*** (0.396)	1.231 (0.876)
match	-0.326 (0.450)	1.959* (1.015)
× information	2.563*** (0.378)	0.373 (0.899)
Baseline Probability $\times 100$	13.52	12.90
Time FE	Yes	Yes
Subject FE	No	Yes
R^2	0.01981	0.74159
N	76,229	9,102

(b) Table 2: Between- and within-subject treatment effects

6.6 Table 3: Willingness to Pay for Carbon Mitigation, Warm Glow, and Fairness

Read:

- Column 1 estimates the baseline model using STANDARD variation and finds WTP of about -0.18 EUR/tCO₂, statistically indistinguishable from zero.
- Column 1 also estimates warm glow utility of about 1.27 EUR per offset, strongly significant.
- Column 2 uses INFORMATION variation and estimates WTP of about 15.99 EUR/tCO₂, strongly significant.
- Column 2 estimates warm glow utility of about 0.74 EUR per offset.
- Column 3 estimates the behavioral model using the panel and finds WTP of 12.84 EUR/tCO₂.
- Column 3 estimates warm glow of 0.86 EUR per offset.
- The fairness term implies that every euro donated by the firm above the consumer's contribution increases the consumer's WTP by about 0.21 EUR.
- The attenuation parameter $\Omega = 0.84$ means within-subject treatment effects are about 84% of between-subject effects.

Detailed interpretation: This table turns the reduced-form treatment effects into preference parameters. The contrast between columns 1 and 2 is stark: the same market can look like pure warm-glow demand or like modest carbon-mitigation demand depending on whether firm participation is salient. Column 3 reconciles this by allowing inattention and fairness preferences. It says consumers do have some WTP for mitigation when impact is processed, but a large part of offset demand is still not about tons of carbon. The fairness multiplier is economically important because firm spending induces additional consumer contributions.

Economic message: The table is the structural core of the paper: revealed WTP is not one fixed number. It depends on attention, framing, and perceived cost sharing between consumers and the firm.

Table 3
Willingness to pay for carbon mitigation, warm glow, and fairness

	(1)	(2)	(3)
	Standard	Information	Entire Sample
WTP in €/tCO ₂ ($-\frac{\lambda}{\eta}$)	-0.18 (5.59)	15.99*** (2.51)	12.84** (5.13)
Warm glow utility in € ($-\frac{\alpha}{\eta}$)	1.27*** (0.24)	0.74*** (0.07)	0.86*** (0.14)
Fairness disutility per € above firm contribution ($-\frac{\phi}{\eta}$)			0.08 (0.16)
Fairness utility per € below firm contribution ($-\frac{\phi}{\eta}$)			0.21*** (0.06)
Price coefficient (ζ)	0.13*** (0.03)	-0.25*** (0.03)	-0.21*** (0.04)
Dynamic Attenuation Effect (Ω)			0.84*** (0.17)
Sample	Buyers' first visit	Buyers' first visit	Buyers in panel
Time & Zip-Code FE	Yes	Yes	Yes
N for between-subject estimation	42440	42186	76229
N for within-subject estimation			9102

Table 3: Willingness to pay for carbon mitigation, warm glow, and fairness

6.7 Table 4: Hypothetical WTP and Figure 6: Email Survey

Read:

- **Table 4** reports an OLS regression of stated WTP in the email survey.
- The baseline stated WTP is 57 cents for 2.4 kg, which translates to 238 EUR/tCO₂.
- The stated WTP number is 1,388% larger than the 16 EUR/tCO₂ revealed-preference estimate from INFORMATION.
- Education about carbon emissions generally has positive but statistically insignificant effects.

- Firm-contribution information does not robustly raise hypothetical WTP in the survey.
- **Figure 6** shows that many consumers think the firm should pay a large share of the offset cost, with a modal response around 50%.
- Figure 6 also shows that stated WTP is higher among supporters of a carbon tax than among opponents.

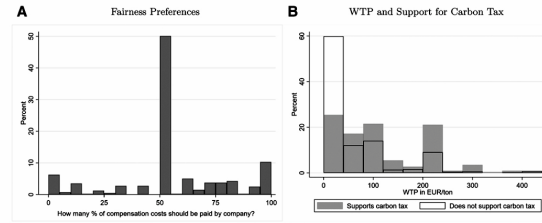
Detailed interpretation: Table 4 is a direct comparison between stated and revealed preferences. The gap is large even though the survey uses the same basic offset setting and small stakes. This implies that hypothetical climate WTP can be much higher than what real market behavior reveals. Figure 6 adds nuance: the survey is not meaningless. People who support carbon taxes report higher WTP, and many people explicitly believe the firm should share the burden. Thus stated WTP contains information about attitudes, but not a reliable dollar value for real willingness to pay in this setting.

Economic message: The survey results caution against using stated climate preferences at face value. They also reinforce the importance of fairness: consumers expect firms to pay part of the cost of carbon mitigation.

Table 4
Hypothetical WTP

	(1) Total WTP (in EUR)
<i>Quantity increase between-subject:</i>	
+4.8kg	6.861 (9.707)
<i>Quantity increase within-subject:</i>	
+2.4kg	31.243*** (2.559)
+4.8kg	32.149*** (3.463)
<i>Between-subject variation in Education and Fairness Treatments:</i>	
2.4kg + Education	7.857 (9.891)
4.8kg + Education	18.314 (12.724)
2.4kg + Firm Contribution	-15.917* (9.056)
4.8kg + Firm Contribution	-1.079 (9.187)
2.4kg + Education & Firm Contribution	7.672 (9.738)
4.8kg + Education & Firm Contribution	-1.216 (9.290)
Constant (baseline offset: 2.4kg)	57.001*** (6.751)
WTP in EUR/tCO ₂	237.50*** (28.13)
R ²	0.0478
N	1,617

(a) Table 4: Hypothetical WTP



(b) Figure 6: Email Survey

7 Result map

- **Experimental design result.** The ten-arm randomization cleanly separates price changes, impact changes, and salient firm contribution.
- **STANDARD reduced form.** Consumers respond to lower prices but not to higher offset quantity; baseline WTP for carbon mitigation is therefore zero.
- **INFORMATION reduced form.** Making firm contribution salient raises price responsiveness and creates positive quantity responsiveness.
- **Belief survey result.** Information does not strongly change beliefs about the environmental harm of a delivery, but may improve perceived effectiveness of larger matches.
- **Structural result.** Standard variation implies zero WTP and large warm glow; information variation implies WTP around 16 EUR/tCO₂; the behavioral model implies WTP around 12.84 EUR/tCO₂, warm glow around 0.86 EUR, and a positive firm-contribution multiplier.
- **Cost-effectiveness result.** Matches are more cost-effective than subsidies because matches increase carbon mitigation for inframarginal buyers.

- **Survey result.** Stated WTP is 238 EUR/tCO₂, far above revealed WTP; survey responses are informative about attitudes but inflated as monetary valuations.

8 Short summary

- Rodemeier estimates WTP for carbon mitigation using real purchases of carbon offsets in an online supermarket.
- The baseline offset compensates 2.4 kg of CO₂ for 24 cents.
- The experiment randomizes price subsidies, quantity matches, and whether consumers are told that the firm pays part of the offset cost.
- In STANDARD, consumers respond to price reductions but not to increased carbon mitigation.
- This implies that standard offset purchases are mostly not driven by intrinsic valuation of the amount of carbon mitigated.
- In INFORMATION, consumers become more price responsive and more impact responsive.
- The main WTP estimate is zero in STANDARD, 16 EUR/tCO₂ in INFORMATION, and 12.84 EUR/tCO₂ in the behavioral model using within-subject variation.
- Warm glow is economically large: consumers receive utility from the act of offsetting independent of the offset's impact.
- Fairness matters: firm contributions increase consumers' willingness to contribute.
- Subsidies are less cost-effective than quantity matches because subsidies transfer money to inframarginal buyers.
- Quantity matches can be more cost-effective because they increase carbon mitigation for all offset buyers.
- The email survey produces stated WTP of 238 EUR/tCO₂, much higher than revealed WTP.
- The paper's main warning is that stated climate preferences can overstate real willingness to pay by an order of magnitude.
- The paper's main managerial lesson is that firms should frame their own sustainable investments transparently and should prefer matches over price subsidies if their goal is carbon mitigation per euro spent.

9 Conclusion

9.1 Core conclusion

The paper shows that voluntary carbon-offset demand is not automatically equivalent to demand for carbon mitigation. In the standard presentation, consumers are willing to buy an offset when its price falls, but they do not buy more when the offset mitigates more carbon. This means the observed purchase can be driven by warm glow or by the symbolic act of offsetting rather than by the quantity of emissions reduced. When the firm makes its own contribution salient, consumers become more attentive to impact and revealed WTP rises, but it remains far below stated survey WTP.

9.2 Economic intuition

The central intuition is that carbon mitigation is cognitively abstract. A grocery delivery creates emissions, but consumers may not have an intuitive sense of what 2.4, 4.8, or 9.6 kg of CO₂ means. A low price is easy to process; an impact quantity is harder. Firm participation changes the psychological frame: consumers see the offset as shared responsibility, may trust it more, and may become more attentive to the scale of mitigation. Thus the same economic object can produce different demand elasticities depending on framing.

9.3 Implications for environmental policy

The results caution against using voluntary offset purchases as a direct measure of the welfare benefits of carbon mitigation. If consumers buy offsets primarily for warm glow, aggregate offset demand may substantially overstate the value consumers place on tons of carbon removed. Conversely, zero scope sensitivity in a low-attention environment does not necessarily mean households have no environmental preferences; it may mean the product interface fails to make impact salient. Policymakers using WTP estimates for cost-benefit analysis should therefore prefer revealed-preference designs with real stakes and explicit impact variation.

9.4 Implications for firms and ESG strategy

The paper implies that corporate sustainability actions can matter for consumer behavior, but the channel is subtle. Firms can increase offset demand by revealing that they pay part of the cost, which activates fairness or shared-responsibility motives. However, if the firm's goal is to maximize carbon mitigated per euro spent, price subsidies are inefficient relative to quantity matches. Matches increase carbon mitigation for all buyers, not just newly induced buyers. This distinction matters for firms that want their climate spending to generate real environmental impact rather than just higher participation rates.

9.5 Implications for stated-preference methods

The email survey highlights a large gap between stated and revealed WTP. The 238 EUR/tCO₂ stated estimate is close to many contingent-valuation estimates in the literature, but far above revealed WTP in the field. This does not make surveys useless: survey WTP is correlated with carbon-tax support and captures attitudes. But the gap warns that hypothetical answers are poor substitutes for real market choices when the goal is monetary valuation.

9.6 Limitations

The offset is low-stakes, the carbon quantity is small, and the setting is a checkout add-on. Results may differ for high-stakes durable goods, energy investments, investment products, or regulatory choices. The offset project is also a particular reforestation project, and consumers' trust in offsets may vary. Finally, INFORMATION may combine multiple mechanisms: fairness, trust, attention, and inferences about firm quality. The paper provides evidence consistent with fairness and attention, but it does not fully decompose every psychological channel.

9.7 Final takeaway

The paper's most important contribution is to show that revealed demand for sustainable products can be behavioral and context-dependent. Consumers can care about acting green, sharing responsibility with firms, and supporting sustainability, while still being insensitive to the actual amount of carbon mitigated. Measuring environmental preferences therefore requires designs that vary price, impact, and information separately.